FARAH ADRIANA BINTI MOHD SAMSUL

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EDUCATION BACKGROUND

Universiti Teknologi Malaysia **2024 - Present**
Master in Philosophy in Electrical Engineering

Universiti Teknologi Malaysia **2019 - 2023**
Bachelor of Engineering (Electrical-Electronics)

PROFESSIONAL DEVELOPMENT

Advanced Semiconductor Academy of Malaysia (ASEM) **April 2026**
National Semiconductor Excellence Program (IC Design)

- **IC Design Fundamentals & ASIC Flow**: RTL Design, Functional Verification, Logic Synthesis, Static Timing Analysis (STA), Floorplanning, Placement & Routing (P&R).
- **EDA Tools**: Synopsys IC Compiler II (ICC II), Design Compiler, PrimeTime, VCS, Zemax, MATLAB.
- **Programming Languages**: Verilog, SystemVerilog, TCL, C.

AI for MY Future Upskilling Program - (2025) **2025**

Train & Place Upskilling Program - (2023) **2023**
Embedded Systems & IoT

WORK EXPERIENCE

Intelligent Circuit Engineering (ICE) **June 2026 - August 2026**
IC Design Trainee

- Designed, integrated, and verified an AMBA AHB-Lite peripheral subsystem with memory-mapped SPI registers using industry-standard simulation flows.
- Developed an AHB Bus Functional Model (BFM) for protocol-compliant read/write verification.
- Built synthesizable RTL and testbenches for AHB-Lite pipelined transfers and peripheral communication.
- Debugged functional and timing issues using simulation waveforms.

Universiti Teknologi Malaysia **October 2024 - August 2026**
Graduate Research Assistant

- Designed and optimized a fiber optic daylighting system to improve indoor natural light distribution and reduce reliance on artificial lighting.
- Conducted ray-tracing simulations using Zemax to model multi-stage concentrator performance, evaluating illuminance distribution, and optimizing system efficiency.
- Iteratively refined designs based on simulation results, achieving measurable improvements in performance and feasibility.
- Processed and analyzed large simulation datasets to identify trends and support data-driven design decisions.

Projek Lebuhraya Usahasama Berhad **February 2024 - October 2024**
Graduate Trainee (Toll Assurance & Recovery)

- Designed and maintained a Power BI dashboard to deliver detailed insights on monthly recovery data to the management.
- Provided administrative support for the Systematic Toll Outstanding Recovery Management (STORM) platform, addressing user inquiries and resolving account-related issues.
- Monitored and recorded daily payment transactions, ensuring accuracy and timely system updates.
- Conducted quarterly compliance audits to verify that plaza operations are aligned with documented working instructions, enhancing compliance and minimizing operational risks.

Medivest Sdn Bhd - Hospital Tuanku Ja'afar **July 2022 - October 2022**
Engineering Trainee (Electrical)

- Managed energy consumption data and monitored retrofit projects to optimize resources.
- Monitored the retrofit project to ensure it was completed on time, within budget, and met all stakeholders' requirements.
- Prepared and presented comprehensive maintenance reports to facility management, highlighting key performance indicators and areas for improvement.

SKILLS

- **IC Design & EDA Tools:** Synopsys IC Compiler II (ICC II), Design Compiler, PrimeTime, VCS, Zemax, MATLAB, AutoCAD
- **Programming & Hardware Languages:** Verilog, SystemVerilog, TCL, C
- **Data Analysis & Visualization:** Power BI, Tableau

LANGUAGES

- **Malay:** Native (Speaking/Writing)
- **English:** Proficient (Speaking/Writing)

REFERENCES

Assoc. Prof. Dr. Ts. Ir Asrul Izam Azmi
Faculty of Electrical Engineering
Universiti Teknologi Malaysia
asrul@utm.my

Student Affairs
Advanced Semiconductor Academy of Malaysia (ASEM)
+603-8320 9499

Certificate of Completion

This certificate is awarded to

Farah Adriana Binti Mohd Samsul ,

I/C:

000208-14-0134

In Recognition of Their Successful Completion of

National Semiconductor Excellence Program
Digital IC Design (Cohort 1)

Organized by
Advanced Semiconductor Academy of Malaysia

Training provider:
Intelligent Circuit Engineering

ANGEL LOW
General Manager,
Advanced Semiconductor
Academy of Malaysia (ASEM)

Y.B. TUAN NG SZE HAN
Selangor Exco of Investment,
Trade and Mobility

B. SREEDHARAN
Senior Application Consultant,
Intelligent Circuit Engineering
(ICE)

Formerly known as: Nusantara